

Teaching Exam

Subject – Computer Science

Topic – Digital Logic



Lecture No.- 1

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Topics to be Covered

Topic



- 1:- Fundamental Of Digital Electronic and Digital Logic
- 2:- signal ✓
- 3:- Digital Signal and Analog Signal ✓
- 4:- Classification Of Number System ✓
- 5:- All four Weighted Type Number System ✓
- 6:- Binary to Decimal Conversion ✓
- 7:- Decimal To Binary Conversion ✓



Topic : Fundamental of Digital Electronics and Logic Design

Digital Electronics and Logic Design are key concepts in both electronics and computer science. Digital systems are at the core of everything from basic devices like calculators to advanced computing systems.

1:- Digital Number systems use binary numbers (0s and 1s) to represent and process information.

2:- Logic gates like AND, OR and NOT are combined to create circuits that perform logical operations on binary inputs.

3:- Combinational circuits produce outputs based only on current inputs, without storing any past information.

4:- Sequential circuits use memory elements, such as flip-flops, to store and process information over time.

The principles of digital electronics and logic design are fundamental in building processors, memory systems and many other computing applications.



What is a Signal?

A physical quantity that has capability to transmit information from one point to another is called a **signal**. Some common **examples** of signals include voltage, current, electromagnetic wave, optical signals, etc.

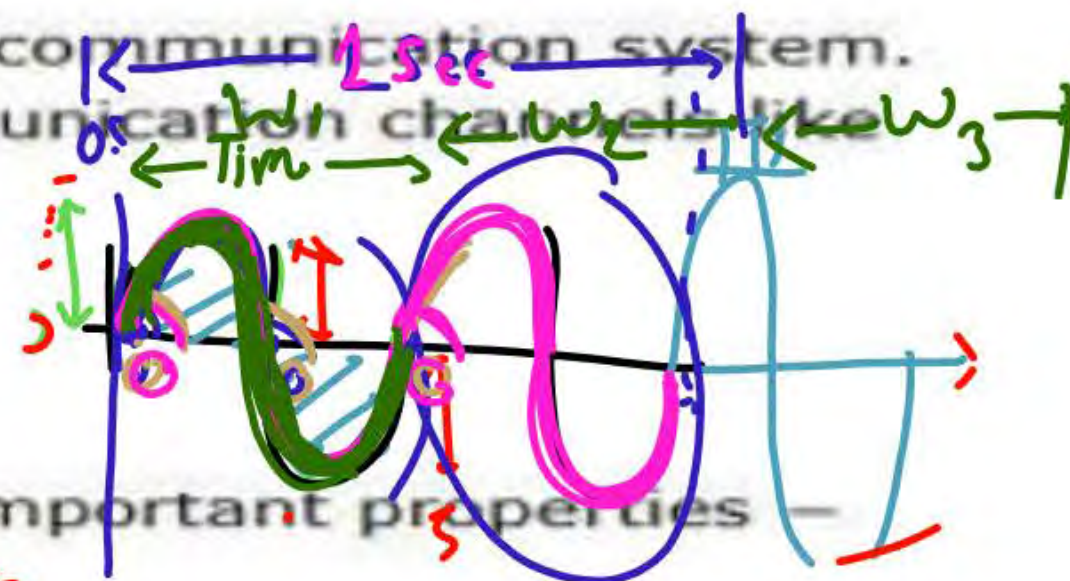
Signals are the backbone of any electronic processing or communication system. These can be transmitted through various types of communication channels like wires, space (electromagnetic waves), optical fibers, etc.

Properties of Signal

In electronics, a signal is characterized by the following important properties –

- **Magnitude** – The intensity or maximum value of a signal is termed as its magnitude.
Handwritten: x_{ax} (pointing to the peak of a sine wave), Amplitude $+5$ and -5 (pointing to the positive and negative peaks of a sine wave), Maximum Amplitude Value (circled in pink).
- **Frequency** – The number of oscillations per second is called frequency of the signal.
Handwritten: $= 2412$ (in pink).
- **Time period** – The time taken to complete one oscillation is called the time period of the signal.
Handwritten: $T = \frac{1}{f}$ (frequency) (in green), Time Period (in green), 2 sec (in pink, pointing to one full cycle of a sine wave).

$$T = \frac{1}{f} \text{ (frequency)}$$





Features of Digital Systems

- Uses Binary Code: Digital systems use binary code, which is a combination of zeros and ones, to represent information.
- Accuracy: Digital systems are more accurate than analog systems because the information is represented in a precise and consistent manner.
- Processing Speed: Digital systems are capable of processing large amounts of data quickly and accurately.
- Noise Immunity: Digital systems are immune to noise and interference, which means that the transmitted information is less likely to be corrupted.

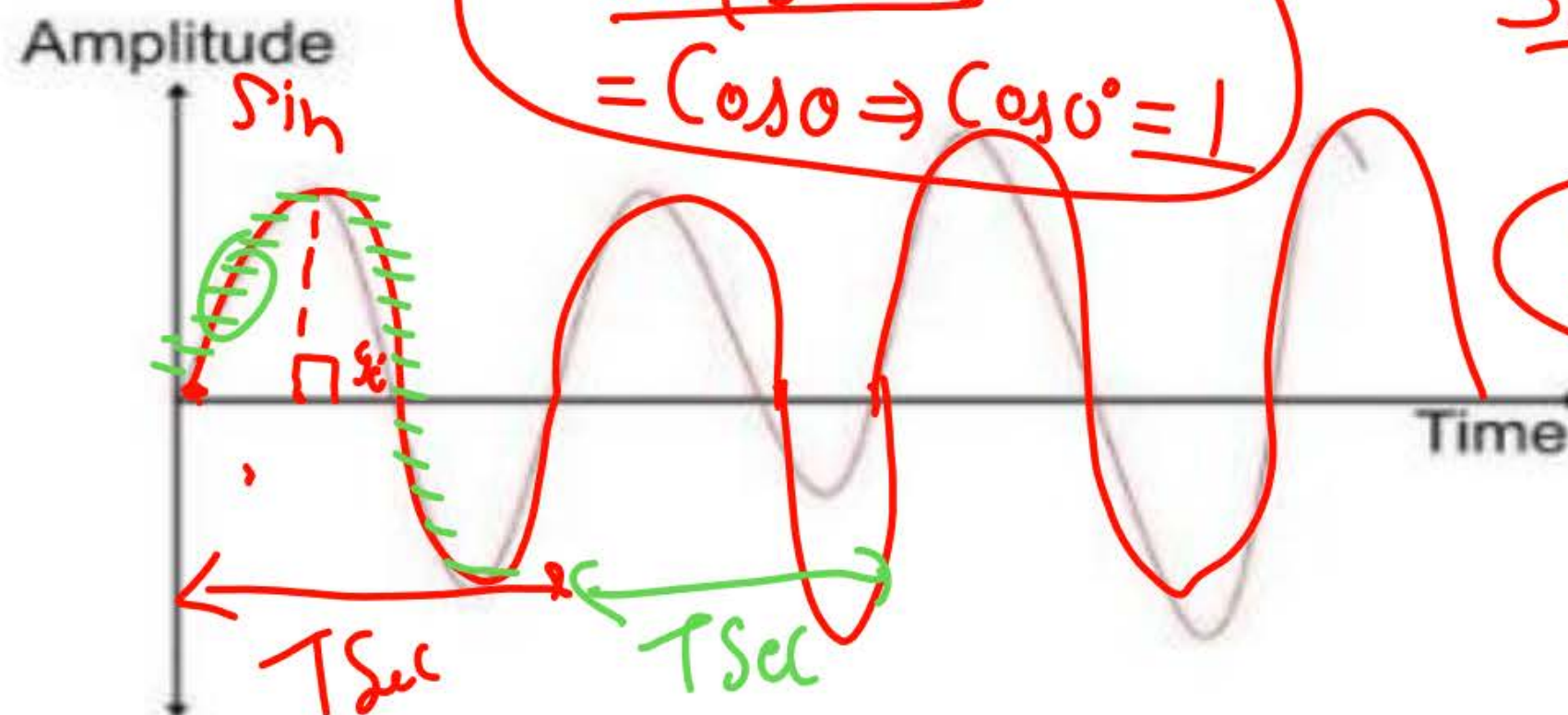
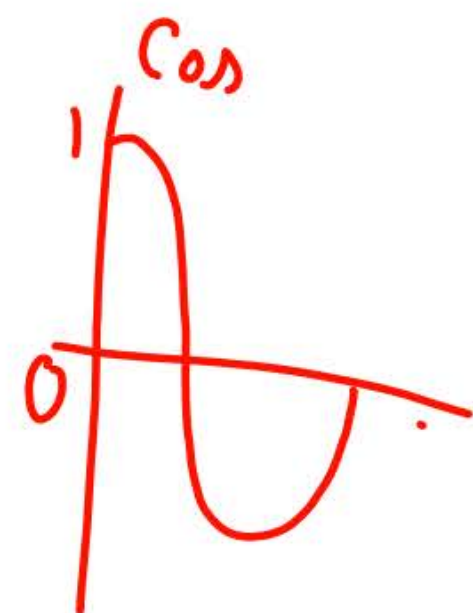


Topic : Analog Signal

$\tan x$

What is Analog System?

Analog system is one that uses continuous time signal or analog signal which is a sinusoidal waveform. Analog system transmits the output in their raw form reducing the time of translation. The amplitude of the signal varies continuously with the time. Analog signals are used to represent sound, temperature, light intensity etc.



$\theta = 0^\circ$
 $\sin(90 + \theta)$
 $= \cos \theta \Rightarrow \cos 0^\circ = 1$

periodic signal
Sinusoidal wave
Sin wave or Cos wave
Phase shift = 90°



Topic : Analog Signal



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Features of Analog Systems

- **Uses Continuous Signals:** Analog systems use continuous signals to represent information, such as electrical voltages or sound waves.
- **Real-World Representation:** Analog systems are better suited for representing real-world phenomena such as sound and light, which are continuous in nature.
- **Smooth Transitions:** Analog systems provide smooth and continuous transitions between different values, which can be important in certain applications such as music or video.
- **Complexity:** Analog systems can be more complex than digital systems due to the need for additional circuitry to process and transmit the signals.

Similarities Between Digital and Analog Systems

- Both can be used to process and transmit information.
- Both can be used in a variety of applications such as audio, video, and telecommunications.
- Both can be used in combination with each other to achieve certain goals, such as using digital signal processing to enhance analog signals.
- Both require some level of circuitry or hardware to function.



Topic : Difference between Digital and Analog Signal



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Difference Between Digital And Analog System

	Analog System	Digital System
Signal	<u>Analog signal</u> represents physical measurements.	<u>Digital signals</u> are discrete and generated by digital modulation.
Waves	<u>Sine Waves</u>	<u>Square Waves</u>
Representation	<u>Continuous range of values to represent information</u>	Uses <u>discrete values to represent information</u>
Technology	Records waveforms as they are.	Samples analog waveforms into a limited set of numbers and then records them.
Data transmissions	<u>Affected by noise during transmission and write/read cycle.</u>	<u>Noise-immune during transmission and write/read cycle.</u>



Topic : Difference between Digital and Analog Signal

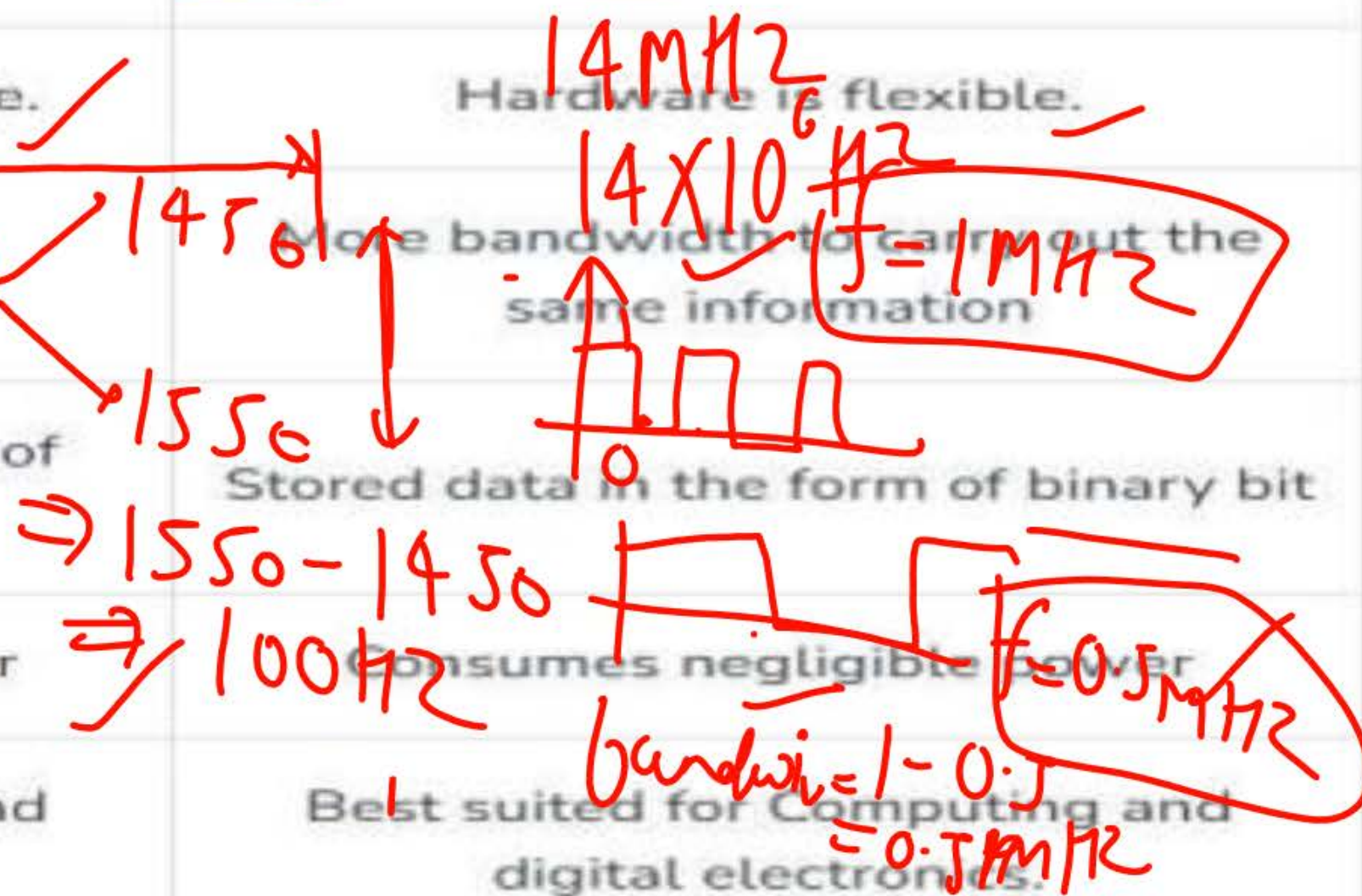
Data transmissions	Affected by noise during transmission and write/read cycle.	Noise-immune during transmission and write/read cycle.
Response to Noise	More likely to get affected	Less likely to get affected
Flexibility	Hardware is not flexible.	Hardware is flexible.
Bandwidth	Less bandwidth	More bandwidth to carry out the same information
Memory	Stored data in the form of wave signal	Stored data in the form of binary bit
Power	Consumes large power	Consumes negligible power
Uses	Best suited for audio and video transmission.	Best suited for Computing and digital electronics.

Spectrum = Range of frequencies

Bandwidth
Width of wave
 $f = 1/T$
wii

frequency width

Spectrum (frequencies)





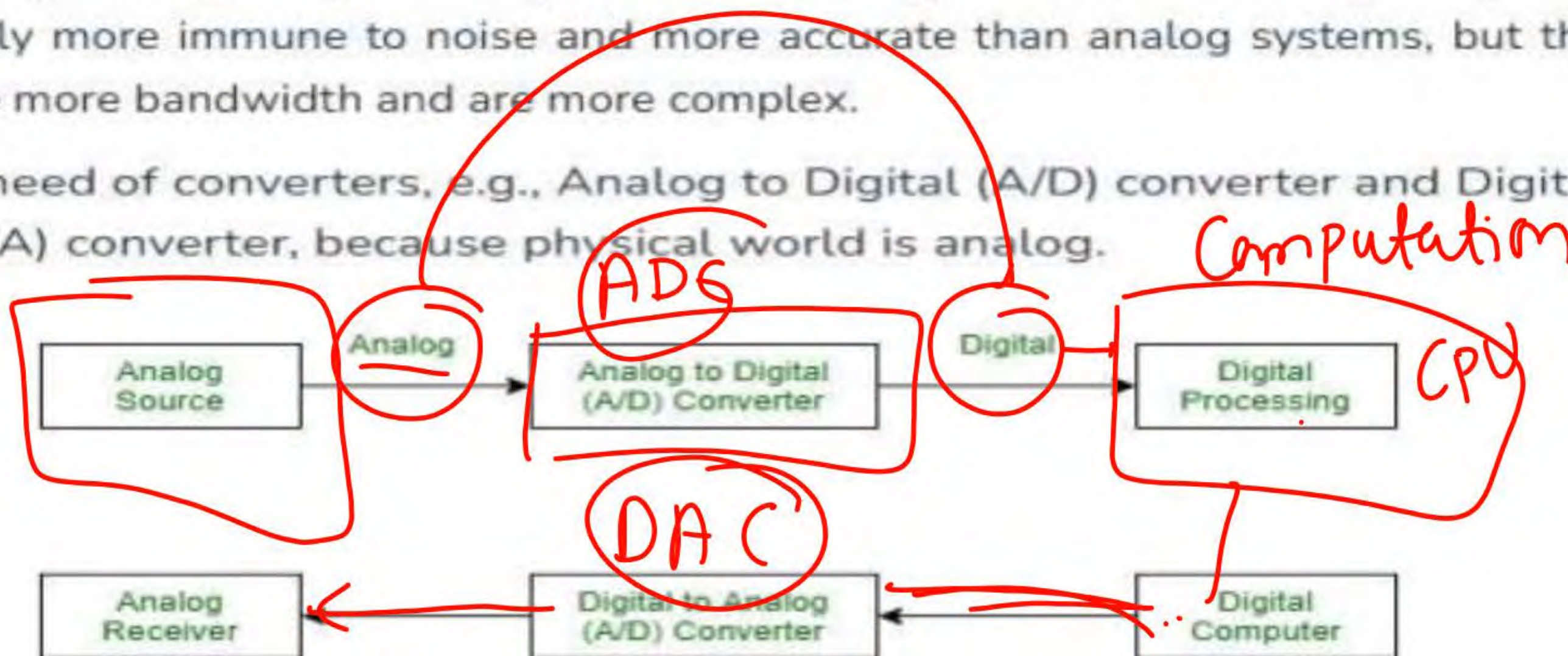
Topic : Conclusion on Digital and Analog System , Convertors

Conclusion

Digital and analog systems differ in how they represent data, noise immunity, bandwidth requirements, complexity, and accuracy. While analog systems represent data using a continuous signal, digital systems represent data using discrete values. Digital systems are generally more immune to noise and more accurate than analog systems, but they also require more bandwidth and are more complex.

There are need of converters, e.g., Analog to Digital (A/D) converter and Digital to Analog (D/A) converter, because physical world is analog.

Real
world
Analog
Signal





Topic : Classification of Number System



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A number is a way to represent arithmetic value, count or measure of a particular quantity. A

number system

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can be considered as a mathematical notation of numbers using a set of digits or symbols. In simpler words the number system is a method of representing numbers. Every number system is identified with the help of its base or radix.

Base or Radix of a Number System:

The base or radix of a number system can be referred as the total number of different symbols which can be used in a particular number system. Radix means "root" in Latin. Base equals to 4 implies there are 4 different symbols in that number system. Similarly, base equals to "x" implies there are "x" different symbols in that number system.

(782)

→ Face Value

10 ⇒ Base Value



Topic : Classification of Number System



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Number System Classification

Positional/Weighted
Number System

- Decimal
- Octal
- Binary
- Hexadecimal
- BCD
- 8-4-2-1 Code

Non-Positional/Non-Weighted
Number System

- Excess-3 Code
- Cyclic Code
- Roma Code
- Gray Code

1 1 e l l 0

@ _ L

1 2 1
↑ ↑ ↑
H T 0

Encoding

1. Positional (or Weighted) Number System:

A positional number system is also known as weighted number system. As the name implies there will be a weight associated with each digit. According to its position of occurrence in the number, each digit is weighted.

2. Non-Positional (or Non-weighted) Number System:

Non-positional number system is also known as non-weighted number system. Digit value is independent of its position. Non-positional number system is used for shift position encodes and error detecting purpose.



Topic : Number System / Positional Number System



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A number system is a way of writing numbers using specific symbols or digits. It helps us represent numbers mathematically. There are different types of number systems, such as the decimal system, binary system, octal system, and hexadecimal system. Here, we will discuss the types of number systems in detail along with solved examples.

Range of hexadecimal N.S $\Rightarrow 0$ to F

$10 = A$
 $11 = B$
 $12 = C$
 $13 = D$
 $14 = E$
 $15 = F$

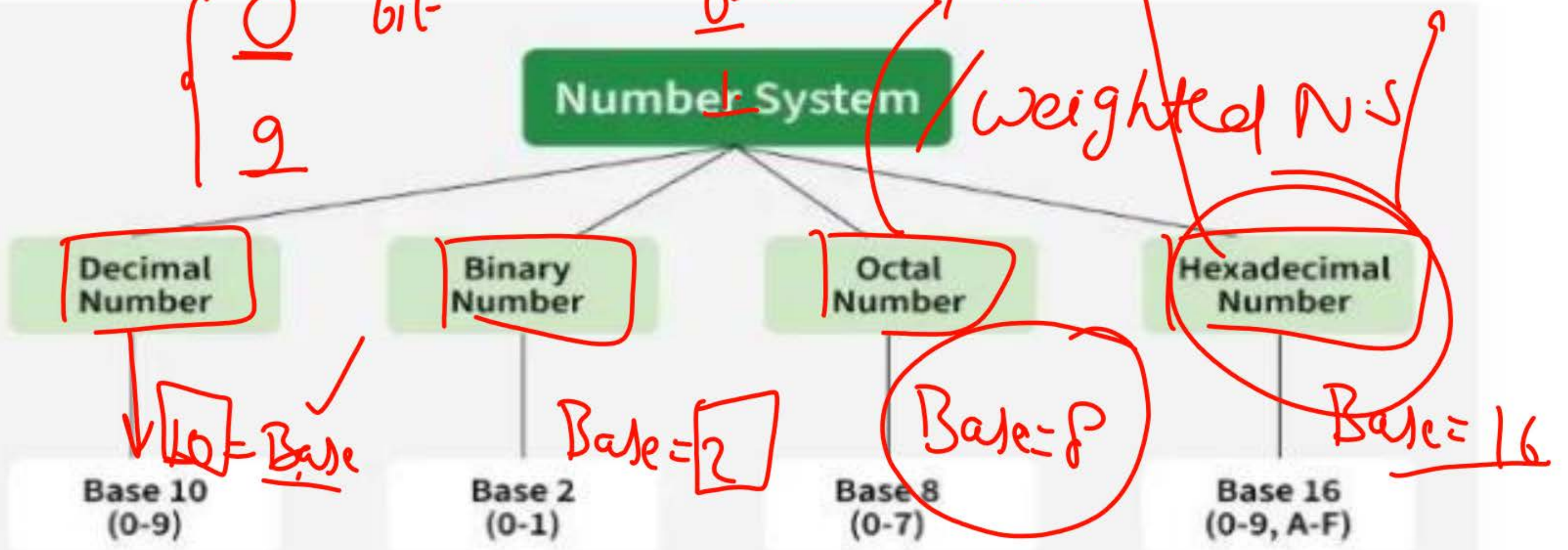
$0 \rightarrow 1$
 $1 \rightarrow 2$
 $2 \rightarrow 3$
 $3 \rightarrow 4$
 $4 \rightarrow 5$
 $5 \rightarrow 6$
 $6 \rightarrow 7$
 $7 \rightarrow 8$
 $8 \rightarrow 9$
 $9 \rightarrow 10$
 $10 \rightarrow 11$
 $11 \rightarrow 12$
 $12 \rightarrow 13$
 $13 \rightarrow 14$
 $14 \rightarrow 15$

Single bit $\Rightarrow 0$

Single bit $\Rightarrow 0$

Single bit $\Rightarrow 0$ to 7

0
 1
 2





Topic : Decimal Number System



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A number system is a method of representing numbers. Every number system is identified with the help of its base or radix. For example, Binary, Octal, Decimal and Hexadecimal Number systems etc. If the **Base value** of a number system is **10**. This is also known as **base-10** or **Decimal number system** which has 10 characters from 0 to 9.

Check whether number is Decimal number system or not ? If yes then find face and base value

- 1:- $(325)_{10} \Rightarrow \text{face value} = 325, \text{Base value} = 10$
- 2:- $(76.25)_{10} \Rightarrow \text{face value} = 76.25, \text{Base value} = 10$
- 3:- $(-325)_{10}$
- 4:- $(268)_8$



Topic : Binary Number System



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Binary Number System uses two digits, '0' and '1', and is the foundation for all modern computing. The word binary is derived from the word "bi," which means two. But what makes it so essential, and how does it work? This article will dive deep into **binary numbers**, binary decimal number conversion and vice versa, 1's and 2's complements, and how they are used in **computer systems**.

There are generally various types of number systems, and among them, the four major ones are,

- Binary Number System (Number system with Base 2)
- Octal Number System (Number system with Base 8)
- Decimal Number System (Number system with Base 10)
- Hexadecimal Number System (Number system with Base 16)

In the Binary Number System, we have a base of 2. The base of the Binary Number System is also called the radix of the number system.

In a binary number system, we represent the number as,

- $(11001)_2$



Check WEATHER Number is Binary or not ?

1:- $(1011)_2$

2:- $(1011)_{10}$

3:- $(1011110072)_2$

4:- $(1000000123)_2$



Octal Number System is a number system with base 8 as it uses eight symbols (or digits) namely 0, 1, 2, 3, 4, 5, 6, and 7. For example, 22_8 , 13_8 , 17_8 , etc. are octal numbers. This number system is mainly used in computer programming as it is a compact way of representing binary numbers with each octal number corresponding to three binary digits.

Check whether Number is Octal Number system:-

1:- $(625)_8$

2:- $(78)_8$

3:- $(795325)_8$

4:- $(10111)_8$



The **Hexadecimal Number System** is a base-16 number system used in diverse fields, especially in computing and digital electronics. It consists of 16 symbols, including numbers 0 to 9 and letters A to F, offering a compact way to represent binary-coded values. The hexadecimal number system is sometimes also represented as 'hex'.

Check whether Number is HexaDecimal Number system:-

1:- $(62FE)_{16}$

2:- $(FGHJ)_{16}$

3:- $(FFFF)_8$

4:- $(10111)_{16}$



How to Write Numbers in Expanded Form?

Go through the below steps to write the numbers in expanded form:

Step 1: Get the standard form of the number.

Step 2: Identify the place value of the given number using the place value chart.

Step 3: Multiply the given digit by its place value and represent the number in the form of (digit $\times$ place value).

Step 4: Finally, represent all the numbers as the sum of (digit $\times$ place value) form, which is the expanded form of the number.



Write down Expanded Form of Decimal Number system

1- $(7648)_{10}$

2:- $(4569.689)_{10}$

3:- $(999999.00789)_{10}$



How to use Binary to Decimal Calculator?

We can easily use the **binary-to-decimal calculator** by following the steps discussed below,

Step 1: Enter the given value in the binary input field.

Step 2: Click on the convert button to convert the binary value into the decimal value.

Step 3: The value shown as the result is the required value in the decimal form.

Binary to Decimal Formula

To convert a **binary number to decimal** we need to perform a multiplication operation on each digit of a binary number from right to left with powers of 2 starting from 0 and add each result to get the **decimal** number of it.

Decimal Number = n^{th} bit $\times 2^{n-1}$

Binary to Decimal Formula

$$n = b_nq + b_{n-1}q^{n-2} + \dots + b_2q^2 + b_1q^1 + b_0q^0 + b_{-1}q^{-1} + b_{-2}q^{-2}$$



Convert Binary To Decimal Number System :-

1:- $(10111000)_2 = (?)_{10}$

2:- $(10101010)_2 = (?)_{10}$

3:- $(0000100001)_2 = (?)_{10}$



How to Convert Decimal to Binary

To convert decimal numbers into binary, there are several methods, including formulas and division techniques. In this explanation, we'll use the remainder method. The steps to convert a decimal number to binary using this method are as follows:

Step 1: Divide the given decimal number by 2, and find the remainder (R_i).

Step 2: Now divide the quotient (Q_i) that is obtained in the above step by 2, and find the remainder.

Step 3: Repeat the above steps 1 and 2 until 0 is obtained as a quotient.

Step 4: Write down the remainder in the following manner: the last remainder is written first, followed by the rest in reverse order ($R_n, R_{(n-1)}, \dots, R_1$). Thus binary conversion of the given decimal number will be obtained.



Convert Decimal to Binary Number System:-

1:- $(789)_{10} = (?)_2$

2:- $(1024)_{10} = (?)_2$

3:- $(9999)_{10} = (?)_2$

THANK YOU